

1. Problem Statement Title & Official Details

Title: Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization

Organization: National Technical Research Organisation (NTRO) | Theme: Blockchain & Cybersecurity | Category: Software

Team ID: 146878 | Team Name: eMitra | Platform Name: Void Vault

2. Background & Strategic National Urgency

With the exponential surge in high-capacity digital storage media across intelligence agencies, defense installations, and state cyber police units, operators face two simultaneous, conflicting operational imperatives:

? Data Sanitization: Securely destroying classified intelligence and citizen records to make recovery impossible.

? Forensic Data Recovery: Recovering deleted, damaged, or anti-forensically concealed evidence during active investigations.

Existing utilities are single-purpose, fragmented, expensive, and foreign-licensed. There is an urgent national need for a unified sovereign platform.

3. The Three Core Mandated Modules

Module 1: Secure Drive Eraser ? Full physical disk sanitization across HDDs, SSDs, NVMe, and USB media adhering to NIST SP 800-88, DoD 5220.22-M, and IEEE 2883-2022. Features readback verification, Shannon entropy mapping, and HPA/DCO hidden area detection.

Module 2: Secure File & Folder Eraser ? Surgical anti-forensic shredding with Alternate Data Stream (ADS) elimination, cluster slack space zeroing, and NTFS Master File Table (MFT) record obfuscation.

Module 3: Advanced File Carving & Recovery ? Commercial-grade carving across formatted and damaged media using signature matching (20+ types), structural AST parsing, Bifragment Gap Carving (BGC), and entropy-based confidence

4. Legal Compliance & Evidentiary Integrity

Full alignment with the Bharatiya Sakshya Adhinyam (BSA) 2023 Section 63 for automated court-admissible forensic certificates, sealed via an immutable append-only SHA-256 Merkle hash chain.